Amicable Numbers

Code:

```
#include <iostream>
#include <vector>
using namespace std;

const int MAXN = 100000;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    vector<int> sumDivisor(MAXN+1, 0);

    for (int divisor=1; divisor <= MAXN/2; divisor++) {
        for (int multiple = divisor * 2;
             multiple <= MAXN;
             multiple += divisor) {
            sumDivisor[multiple] += divisor;
        }
    }

    vector<pair<int, int>> amicablePairs;

    for (int a = 2; a <= MAXN; a++) {
        int b = sumDivisor[a];

        if (b > a && b <= MAXN && sumDivisor[b] == a) {
            amicablePairs.push_back({a, b});
        }
    }
}
```

Scanned with CamScanner

```
int L, R;

while (cin >> L >> R) {
    if (L == 0 && R == 0) {
        break;
    }
    int answer = 0;

    for (const auto &p : amicablePairs) {
        if (p.first >= L && p.second <= R) {
            answer++;
        }
    }
    cout << answer << "\n";
}
return 0;
```

Input:

```
1 300
200 300
1 1000
1 10000
0 0
```

Output:

```
1
1
1
5
```